

2. Frontend Technologies

Recommended

Framework

- [Next.js](#)

Why?

- [SEO support](#)
- [Fast rendering](#)
- [Easy API integration](#)
- [Modern UI](#)

Styling

- [Tailwind CSS](#)

UI Components

- [shadcn/ui](#)

Provides:

- [Chat UI](#)
- [Cards](#)
- [Dialogs](#)
- [Tables](#)

State Management

- [Zustand](#)

Frontend Dependencies

[npm install axios](#)

[npm install zustand](#)

[npm install react-markdown](#)

[npm install tailwindcss](#)

[npm install shadcn-ui](#)

3. Backend Technologies

Framework

FastAPI

Why?

- Extremely fast
- Async support
- AI-friendly
- Swagger docs

Install:

pip install fastapi
pip install uvicorn

4. AI Framework

Best Choice

LangGraph

Why?

Traditional AI:

User

↓

One Agent

↓

Response

LangGraph:

User

↓

Planner

↓

Research Agent

↓

Fact Agent

↓

Summarizer

↓

Final Answer

Install:

pip install langgraph

5. LLM Options

Free Local

Ollama

Models:

- Llama 3

- [Mistral](#)
- [Gemma](#)

Install:

ollama pull llama3

Cloud

- [OpenAI Platform](#)
- [Anthropic Claude API](#)
- [Google AI Studio](#)

6. RAG System

RAG = Retrieval Augmented Generation

Instead of relying only on model memory:

Question

↓

Search Documents

↓

Retrieve Relevant Chunks

↓

Send to LLM

↓

Accurate Answer

Libraries:

pip install langchain

pip install sentence-transformers

7. Embedding Models

Convert text into vectors.

Recommended:

Sentence Transformers

Example:

all-MiniLM-L6-v2

8. Vector Database

Stores document embeddings.

Beginner

Chroma

Install:

pip install chromadb

Production

Qdrant

or

Pinecone

9. Relational Database

Stores:

- Users
- Chats
- Reports
- History

Recommended:

PostgreSQL

Install:

pip install psycopg2

ORM:

SQLAlchemy

10. Search Capabilities

Research Agent should search:

- PDFs
- Websites
- Research papers

Libraries:

pip install pypdf

pip install beautifulsoup4

pip install requests

11. Research Paper Search

Use:

[arXiv API Documentation](#)

For academic papers.

12. Agent Design

Agent 1: Planner

Input:

Explain Quantum Computing

Output:

1. Search papers
2. Gather information
3. Verify sources
4. Summarize

Agent 2: Research Agent

Responsibilities:

- Web search
- PDF search
- arXiv search

Agent 3: Fact Checker

Checks:

- Sources
- Contradictions
- Hallucinations

Agent 4: Summarizer

Creates:

- Notes
- Bullet points
- Key concepts

Agent 5: Report Generator

Produces:

Executive Summary

Findings

References

Conclusion

13. Suggested Database Schema

Users

users

id

name

email

password

Chats

chats

id

user id

query

response

timestamp

Documents

documents

id

title

source

embedding_id

Reports

reports

id

user id

title

content